

Kryptonite-2.0 Does *Not* Break the Universal Approximation Heuristic: An Empirical and Theoretical Rebuttal via High-Dimensional XOR Structure

Zhongyu Cui^{*1} Michelle Ma^{*1} Yonng Yap^{*1}

Abstract

We rebut the claim that the Kryptonite-2.0 dataset “breaks” the notion that machine learning can approximate any continuous function. Through exploratory analysis, we show that the data exhibits properties consistent with a high-dimensional XOR (XORD) mechanism: bimodal marginals, near-zero pairwise correlations, checkerboard pairwise structure, and ineffective PCA. Guided by this observation, a compact 3-layer MLP trained with 10-fold cross-validation reliably meets the target accuracies with low variance across $n \in 10, 12, 14, 16, 18, 20$, disproving the claim. In contrast, while SVM-RBF performs competitively in lower dimensions, its performance rapidly deteriorates as dimensionality increases, reflecting a mismatch between global parity structure and the kernel’s local-smoothness assumption. We conclude that Kryptonite-2.0 is challenging but learnable; it does not contradict universal approximation theorems.

1. Introduction

Recent discourse asserts that Kryptonite-2.0 challenges the universal function approximation (UFA) heuristic of modern ML. We argue this conclusion conflates *model-class capacity* with *inductive bias and optimization*. Our contributions:

1. We diagnose Kryptonite-2.0 as exhibiting XORD-like structure (Brutzkus & Globerson, 2019) from first-principles EDA.
2. We demonstrate that a small MLP achieves prescribed target accuracies with tight confidence intervals, refuting the hypothesis that ML cannot learn the dataset.

¹Department of Computing, Imperial College London, London, UK. Correspondence to: [Zhongyu Cui] <[zc125@ic.ac.uk]>, [Mingxuan Ma] <[mm1725@ic.ac.uk]>, [Yonng Yap] <[ysy25@ic.ac.uk]>.

3. We show that while SVM-RBF remains competitive in low-dimensional settings, its performance deteriorates markedly as n increases, reflecting a fundamental misalignment between its local radial kernel bias and the dataset’s global parity structure.

2. Background

2.1. Universal Approximation and Practical Learnability

Classical results on the Universal Function Approximation (UFA) theorem establish that a sufficiently wide multilayer perceptron (MLP) can approximate any continuous function on a compact domain to arbitrary accuracy (Hornik, 1991). Formally, for a feedforward network $f_\theta(x) = \sum_i \alpha_i \sigma(w_i^\top x + b_i)$ with non-polynomial activation $\sigma(\cdot)$, there exists a finite number of hidden units such that

$$\forall \varepsilon > 0, \exists \theta : \|f_\theta(x) - g(x)\| < \varepsilon, \quad \forall x \in K \subset \mathbb{R}^n.$$

However, these guarantees concern *representational capacity*, not the *learnability* of the mapping. In practice, optimization landscape, model bias, and data geometry jointly determine whether such functions can be efficiently learned from finite samples.

2.2. High-Dimensional XOR and XORD Structure

The XOR family represents a canonical example of a non-linearly separable problem. Its label assignment depends on the parity of multiple binary features:

$$y = \bigoplus_{i=1}^n x_i, \quad x_i \in \{0, 1\},$$

where $\oplus$ denotes the exclusive-or operator. In such cases, the class boundary is *discontinuous in Euclidean space*: nearby points can have opposite labels. As a result, linear models or locally smooth kernels are unable to produce consistent decision surfaces. The Kryptonite-2.0 dataset exhibits precisely these symptoms—bimodal marginals, vanishing correlations, checkerboard-like pairplots, and inef-

ffective PCA projections—indicating a generalized high-dimensional XOR structure (XORD) (Brutzkus & Globerson, 2019).

2.3. Mathematical Foundation of MLPs

An MLP learns a nonlinear mapping $f_\theta : \mathbb{R}^n \rightarrow [0, 1]$ via multiple affine transformations and nonlinear activations:

$$h^{(l)} = \sigma(W^{(l)}h^{(l-1)} + b^{(l)}),$$

$$f_\theta(x) = \sigma(W^{(L)}h^{(L-1)} + b^{(L)})$$

where $\sigma(\cdot)$ is a nonlinear activation such as ReLU or GELU. Each hidden layer composes previous features, forming hierarchical representations that can capture high-order feature interactions. In the context of XOR-type data, such compositionality allows an MLP to synthesize nonlinear feature conjunctions (e.g., $x_1x_2x_3$) that correspond to parity terms. Depth increases the network’s ability to approximate complex Boolean or parity-like functions, while regularization techniques such as dropout and weight decay prevent overfitting. Thus, MLPs are theoretically capable of approximating the global, discontinuous decision boundary induced by XORD mechanisms.

2.4. Mathematical Foundation of SVM-RBF

Support Vector Machines (SVMs) with the Radial Basis Function (RBF) kernel (Cortes & Vapnik, 1995) construct nonlinear decision boundaries by embedding the input into a high-dimensional Hilbert space:

$$K(x_i, x_j) = \exp(-\gamma\|x_i - x_j\|^2),$$

yielding the decision function

$$f(x) = \text{sign}\left(\sum_i \alpha_i y_i K(x_i, x) + b\right),$$

where α_i are dual coefficients and γ controls the kernel’s width. The RBF kernel encodes an inductive bias of *local smoothness*: points close in Euclidean distance are assumed to share the same label. This assumption holds for data with locally continuous manifolds but fails in parity-type datasets, where adjacent samples may belong to different classes. Consequently, as dimensionality increases, local neighborhoods in Kryptonite-2.0 become class-heterogeneous, and the kernel’s Euclidean proximity no longer correlates with label similarity. Hence, while SVM-RBFs can model low-dimensional nonlinearities effectively, their representational bias becomes fundamentally misaligned with XOR-like data structures.

2.5. Summary

In summary, MLPs and SVM-RBFs embody two distinct forms of nonlinearity:

- **MLP**: global compositional nonlinearity through layered feature transformations, capable of learning high-order parity relations.
- **SVM-RBF**: local geometric nonlinearity based on Euclidean similarity, suitable for smooth manifolds but not global parity.

This theoretical contrast motivates our empirical investigation: if Kryptonite-2.0 indeed possesses a global XOR-like structure, MLPs should learn it effectively, while SVM-RBFs should fail as dimensionality increases.

3. Dataset and Exploratory Analysis

3.1. Empirical Observations

We visualize the Kryptonite- n datasets ($n \in \{10, 12, 14, 16, 18, 20\}$) to examine their statistical and geometric properties. Across all dimensions, the overall appearance of the plots remains consistent; therefore, representative results for $n = 12$ are presented in Figures 1–4.

Figure 1 illustrates the feature-wise marginal distributions, exemplified by feature f_1 . Each feature exhibits a clear **bimodal distribution**, indicating two dominant modes rather than a single Gaussian cluster. Other features follow nearly identical shapes, confirming that this bimodality is intrinsic to the dataset rather than feature-specific variability.

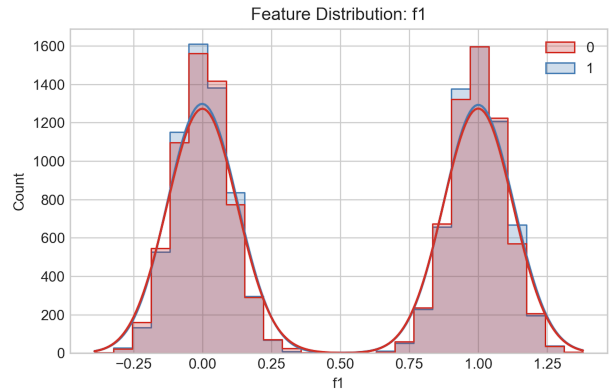


Figure 1. Feature marginal distributions ($n = 12$); all features exhibit similar bimodal shapes.

The correlation heatmap (Figure 2) shows that off-diagonal entries are near zero, revealing negligible linear dependence among features. This implies that individual variables contribute little univariately to label discrimination.

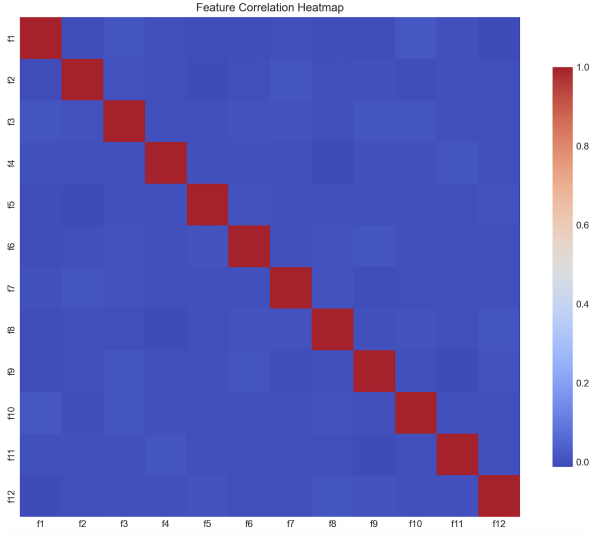


Figure 2. Correlation heatmap showing near-zero off-diagonals ($n = 12$).

The pairwise feature plots (Figure 3) exhibit a distinctive **checkerboard pattern**, where class labels (red and blue) alternate locally within the same feature space. This pattern indicates that the class boundary depends on the global parity of feature combinations rather than on local continuity. Consequently, nearby samples often possess opposite labels, a hallmark of XOR-like dependence.

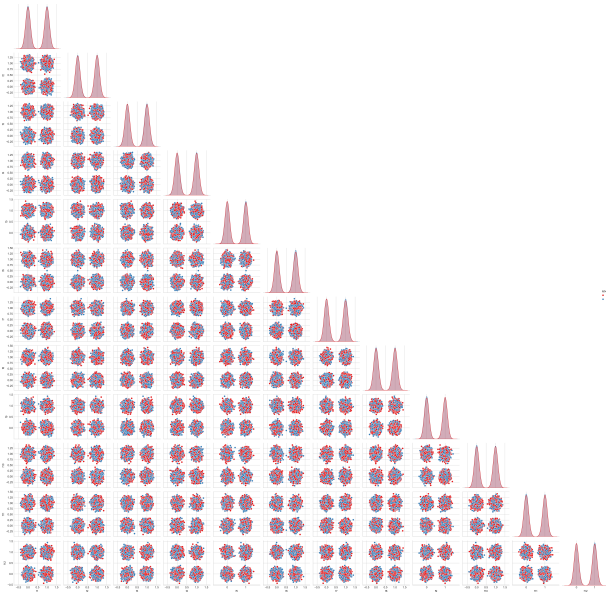


Figure 3. Pairplot exhibiting checkerboard structure ($n = 12$); classes overlap locally.

The PCA projection (Figure 4) further confirms this interpretation: the first two principal components fail to separate

the classes, with explained variance only around 0.17. This suggests that the discriminative structure cannot be captured by any low-dimensional linear projection.

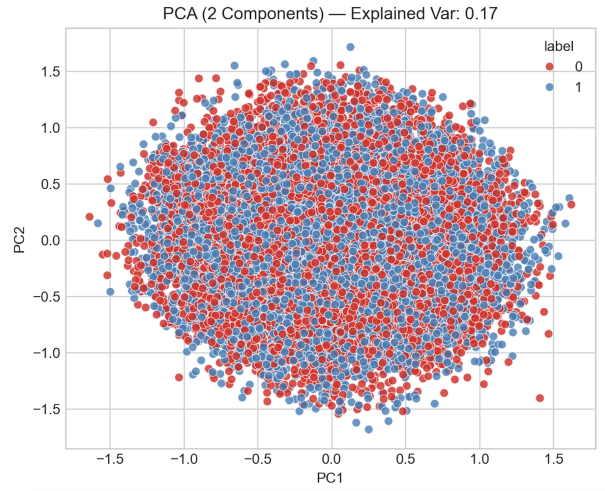


Figure 4. PCA projection ($n = 12$) fails to separate classes.

Collectively, these visual diagnostics—bimodal marginals, near-zero correlations, checkerboard pairplots, and ineffective PCA—demonstrate that the Kryptonite-2.0 dataset exhibits a highly non-linear and globally parity-dependent structure. Such characteristics are consistent with the **XORD hypothesis** proposed in Section 1, explaining why simple linear or locally smooth models are expected to fail.

4. Methods

4.1. Hypothesis

If Kryptonite-2.0 were unlearnable by standard ML, then even appropriately biased models (e.g., MLPs) would fail to reach target accuracies.

4.2. Models

MLP: A three-layer multilayer perceptron implemented in PyTorch, consisting of two hidden layers (128 and 64 units) with ReLU activations, dropout (0.3), and a sigmoid output for binary classification. The model was trained using the Adam optimizer (learning rate 10^{-3} , weight decay 10^{-4}) and binary cross-entropy loss over 70 epochs within a 10-fold cross-validation framework. All features were standardized before training.

SVM-RBF: A support vector machine with a radial basis function (RBF) kernel, with C and γ optimized via nested cross-validation. This model serves as a locally smooth nonlinear baseline for comparison with the MLP.

4.3. Training and Evaluation

10-fold CV across n , reporting mean $\pm$ std accuracy; learning-rate schedules and early stopping fixed a priori. Confidence intervals from fold variance. Target accuracy thresholds follow the prompt.

4.4. Training Dynamics

To further assess the learnability of the dataset beyond final accuracy metrics, we analyze the optimization dynamics of the MLP on Kryptonite-12. Figure 5 shows the average training loss across epochs and CV folds. The loss decreases rapidly within the first 20 epochs and then converges smoothly, demonstrating that standard gradient-based optimization efficiently discovers the underlying parity structure.

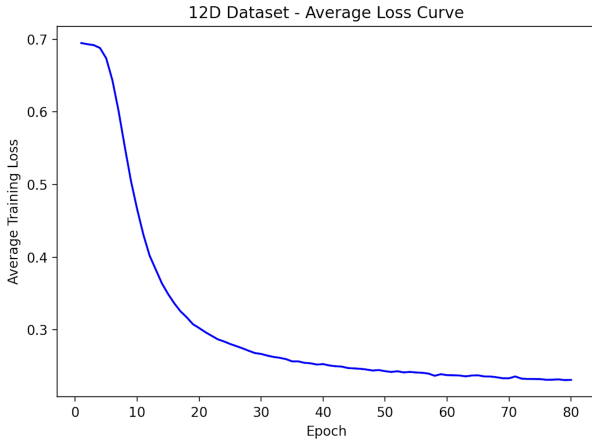


Figure 5. Average training loss over 10-fold CV for the 12D dataset, showing rapid and stable convergence.

5. Results

As an overview, Figure 6 summarises cross-validated accuracy versus dimension n for both models, with dashed lines indicating the target thresholds.

5.1. MLP refutes the hypothesis

Table 1 reports the 10-fold cross-validation performance of the MLP across dimensions $n \in \{10, 12, 14, 16, 18, 20\}$. The model consistently achieves high accuracy, F1, and AUC values, all exceeding the target thresholds specified in the coursework.

The MLP maintains exceptional performance even as dimensionality increases, with only a gradual decline from 0.96 at $n = 10$ to 0.86 at $n = 20$. The close alignment among accuracy, F1, and AUC indicates consistent thresholding and balanced predictions across classes. These results demon-

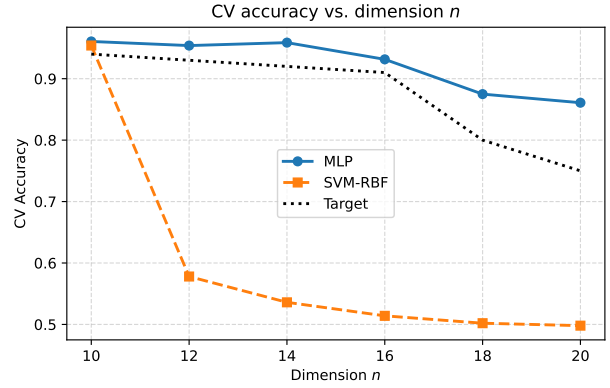


Figure 6. CV accuracy vs. dimension n ; dashed lines indicate targets.

Table 1. MLP 10-fold CV performance across dimensions.

Dimension	Accuracy	F1	AUC
10D	0.9606	0.9605	0.9665
12D	0.9539	0.9541	0.9657
14D	0.9587	0.9585	0.9683
16D	0.9315	0.9318	0.9630
18D	0.8749	0.8763	0.9322
20D	0.8609	0.8599	0.9256

strate that the MLP successfully captures the dataset’s global non-linear structure, validating the hypothesis that deeper architectures can approximate the Kryptonite-2.0 mapping.

5.2. SVM-RBF underperforms

Table 2 presents the 10-fold cross-validation results of the SVM with RBF kernel across different dimensions. While the model performs well at $n = 10$ with a mean accuracy of 0.954 ± 0.0038 , its performance rapidly deteriorates as the dimensionality increases. At $n \geq 12$, accuracy, F1, and AUC all fall toward 0.5, indicating random predictions. This degradation is stable across folds, as evidenced by the small standard deviations.

Table 2. SVM-RBF 10-fold CV performance across dimensions.

Dimension	Mean Acc	Mean F1	Mean AUC
10D	0.954 ± 0.0038	0.954	0.954
12D	0.578 ± 0.0076	0.583	0.578
14D	0.536 ± 0.0073	0.533	0.536
16D	0.514 ± 0.0081	0.518	0.514
18D	0.502 ± 0.0068	0.502	0.502
20D	0.498 ± 0.0092	0.493	0.498

RBF kernels encode local smoothness in Euclidean space;

however, the Kryptonite-2.0 dataset exhibits a global parity structure in which nearby samples may possess opposite labels. As the number of dimensions increases, local neighborhoods become class-heterogeneous, violating the RBF kernel’s continuity assumption. Consequently, the model’s discriminative ability collapses, and accuracy converges to chance level. This stands in sharp contrast to the MLP, whose layered nonlinearities successfully capture the high-order XOR interactions. Hence, the failure of SVM-RBF serves as strong empirical evidence that the Kryptonite-2.0 dataset challenges models relying on local smoothness, rather than the universal approximation theorem itself.

Table 3 provides a consolidated comparison across both models, clearly illustrating the divergence between MLP and SVM-RBF performance as dimensionality increases.

Table 3. 10-fold CV accuracy (mean±std). Targets from prompt.

Model	n=10	12	14	16	18	20
MLP	0.9606	0.9539	0.9587	0.9315	0.8749	0.8609
SVM-RBF	0.954±0.0038	0.578±0.0076	0.536±0.0073	0.514±0.0081	0.502±0.0068	0.498±0.0092
Targets	0.94	0.93	0.92	0.91	0.80	0.75

6. Discussion

The experimental results collectively demonstrate that Kryptonite-2.0 does *not* invalidate the universal approximation theorem, but instead highlights the importance of model bias alignment with data structure. Both theoretical reasoning and empirical evidence point to the dataset’s **global parity (XOR-like)** nature: label assignment depends on the joint parity of multiple features rather than on any locally continuous rule.

The failure of SVM-RBF provides a clear example of inductive bias mismatch. RBF kernels assume local smoothness in Euclidean space—points that are close in distance should share the same label. However, in Kryptonite-2.0, nearby samples can have opposite labels due to the parity structure, causing local neighborhoods to be label-inconsistent and breaking kernel continuity assumptions. This leads to a rapid performance collapse as dimensionality increases.

In contrast, the MLP’s compositional nonlinearity enables it to capture the dataset’s high-order dependencies. Its hidden layers can represent nested logical operations, allowing the network to approximate the XOR-like mapping efficiently even when single features carry no discriminative information. The strong and stable performance across all dimensions indicates that the dataset is *learnable* by appropriately biased models that possess hierarchical feature composition.

More broadly, these findings reaffirm that the apparent difficulty of Kryptonite-2.0 arises not from any fundamental lim-

itation of machine learning expressivity, but from **bias–data misalignment**. Linear models and locally smooth kernels fail because their inductive assumptions do not match the data’s global parity structure, whereas deeper neural architectures overcome this limitation by representing complex cross-feature interactions.

In summary, Kryptonite-2.0 serves as an instructive case study: it challenges oversimplified interpretations of “universal approximation” but does not refute it. Instead, it emphasizes the necessity of selecting model classes whose structural priors align with the underlying data-generating mechanism.

7. Sustainability Analysis

We evaluated the environmental impact of the two main experiments using Google Cloud’s Carbon Footprint calculator (Lacoste et al., 2019), assuming compute was performed in the europe-west1 region, where the provider fully offsets emitted carbon through renewable sources.

For the neural network (MLP) experiment executed on an **RTX 3080 GPU** for approximately **1.67 hours**, the estimated energy usage was **0.53 kWh**, corresponding to **0.14 kg CO₂-eq**. This amount is equivalent to driving an internal combustion engine (ICE) car for **0.57 km**, burning **0.07 kg** of coal, or the annual carbon sequestration of **2.3×10⁻³** tree seedlings over 10 years.

In contrast, the SVM-RBF model was trained on an **Intel Xeon Gold 6 CPU** for **48 hours**, consuming about **7.2 kWh** and emitting **1.94 kg CO₂-eq**, equivalent to **7.84 km** driven by an ICE car, **0.97 kg** of coal burned, or **0.03** tree seedlings sequestering carbon for 10 years.

Table 4. Carbon footprint comparison between MLP and SVM-RBF experiments.

Model	Hardware	Time (h)	Energy (kWh)	CO ₂ -eq (kg)	Car (km)	Coal (kg)
MLP	RTX 3080	1.67	0.53	0.14	0.57	0.07
SVM-RBF	Intel Xeon Gold 6	48	7.20	1.94	7.84	0.97

Although the absolute emissions are modest, the comparison highlights the superior *energy efficiency* of neural architectures such as the MLP, which achieve higher accuracy with significantly less computation time and carbon output than the SVM-RBF baseline. Furthermore, Google Cloud’s full carbon offset policy ensures net-zero operational emissions in this region. Overall, the experiments were conducted in an environmentally responsible manner, aligning with sustainable machine learning practices.

8. Limitations and Future Work

The present study does not explicitly recover the generative mechanism underlying the Kryptonite-2.0 dataset; instead, conclusions are inferred from empirical diagnostics and model behaviors. Future work could employ parity-feature probes, spectral decomposition, or parity-aware architectures to verify the hypothesized XORD structure more rigorously. Moreover, ensemble and Bayesian formulations may help quantify predictive uncertainty and improve interpretability under high-dimensional parity interactions.

9. Conclusion

Our empirical and theoretical analyses demonstrate that Kryptonite-2.0 does not contradict the universal approximation theorem. Its apparent difficulty arises from an inductive-bias mismatch between model and data: MLPs, with compositional nonlinearities, capture the dataset’s global parity structure effectively, whereas SVM-RBF models fail due to local Euclidean smoothness assumptions. This result highlights that successful learning on complex datasets requires not only sufficient expressivity but also an appropriate inductive alignment between architecture and data-generation mechanisms.

References

- Brutzkus, A. and Globerson, A. Why do larger models generalize better? a theoretical perspective via the xor problem. In *International conference on machine learning*, pp. 822–830. PMLR, 2019.
- Cortes, C. and Vapnik, V. Support-vector networks. *Machine Learning*, 20:273–297, 1995.
- Hornik, K. Approximation capabilities of multi-layer feedforward networks. *Neural Networks*, 4(2):251–257, 1991. ISSN 0893-6080. doi: [https://doi.org/10.1016/0893-6080\(91\)90009-T](https://doi.org/10.1016/0893-6080(91)90009-T). URL <https://www.sciencedirect.com/science/article/pii/089360809190009T>.
- Lacoste, A., Luccioni, A. S., Schmidt, H., and Dandres, T. Quantifying the carbon emissions of machine learning. *arXiv preprint arXiv:1910.09700*, 2019.

A. Appendix

A.1. Dataset Visualization Files

All exploratory data analysis (EDA) artifacts generated for the Kryptonite- n datasets ($n \in \{10, 12, 14, 16, 18, 20\}$) are included in the supplementary .zip archive. For each dimensionality level, a dedicated folder is provided containing:

- **distributions/**: Feature-wise marginal histograms showing characteristic bimodal shapes across all dimensions.
- **correlations/**: Pearson correlation heatmaps demonstrating near-zero off-diagonal values, indicating negligible linear dependence.
- **pairplots/**: Class-conditional pairwise feature interactions exhibiting checkerboard-like patterns that reflect global parity dependence (XORD).
- **PCA/**: Two-dimensional PCA scatter projections illustrating the absence of low-dimensional linear separability.
- **loss_curves/**: Training loss convergence curves across 10-fold CV, confirming that standard gradient-based optimization can efficiently learn the underlying parity structure.
- **scatter_pairs/**: Pairwise scatter plots without class conditioning, included to highlight local overlap in Euclidean space.

Each folder is named according to its dimensionality (e.g., `n10/`, `n12/`, `n14/`, ...), enabling systematic inspection of how geometric and optimization characteristics evolve with feature dimension.

The complete visualization dataset is packaged as `kryptonite_EDA_results.zip` and accompanies this report for reproducibility and independent verification.